

Lucas Castillo



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Education

- 2021 – 2025 **Ph.D. in Psychology** UNIVERSITY OF WARWICK, UK.
Supervisor: Prof. Adam Sanborn
Thesis title: *A Local Sampling Account of Random Generation* (📄)
- Jul-Sep 2024 **Research visit** at UNIVERSITY OF HAMBURG, Germany
under the supervision of Prof. Sebastian Gluth.
- 2019 – 2020 **M.Sc. Cognitive and Decision Sciences** UNIVERSITY COLLEGE LONDON, UK
Dissertation title: *Does newly learned causal knowledge guide the perception of temporal order?*
- 2012 – 2016 **B.Sc. Psychology** UNIVERSIDAD DE SEVILLA, Spain.

Employment

- January 2026 - Present **Research Fellow**, UNIVERSITY OF WARWICK, United Kingdom.
ESRC-funded project developing a single cognitive model that explains behaviour across several tasks.
- July – September 2024 **Predoctoral Researcher**, UNIVERSITY OF HAMBURG, Germany.
Carried out research project under the supervision of Prof. Sebastian Gluth.
- 2023 – 2024 **Research Assistant**, UNIVERSITY OF WARWICK, United Kingdom.
Continued development of *samplr* **R Package**.
- 2021 – 2023 **Teaching Assistant**, UNIVERSITY OF WARWICK, United Kingdom.
Assisted teaching of two **statistics modules** (frequentist and Bayesian) at MSc level.
- 2020 – 2021 **Research Assistant**, UNIVERSITY OF WARWICK, United Kingdom.
Started the *samplr* **R Package**, developed **lab website**, and carried out own research.

Acquired Funding

- 2025 **Enhancing Research Fund (£2860)** UNIVERSITY OF WARWICK, UK
(Co-applicant). Purpose: Organizing open research workshops aimed at Early Career Researchers.
- Behaviour Spotlight Small Grant Scheme (£2000)** UNIVERSITY OF WARWICK, UK
(Co-applicant). Project: *Rumination and Random Generation*
- 2024 **Behaviour Spotlight Small Grant Scheme (£2000)** UNIVERSITY OF WARWICK, UK
(Co-applicant). Project: *Exploration Behaviour in Decisions from Experience*

Publications

Journal Articles

- 4 **Castillo, L.,** León-Villagrà, P., Falbén, J. K., Chater, N., & Sanborn, A. N. (2026). *Random Generation is What Comes to Mind in Naturalistic Settings*. *Cognition*, 274, 106554.
🔗 <https://doi.org/10.1016/j.cognition.2026.106554>

Publications (continued)

- 3 Spicer, J., Li, Y.-X., **Castillo, L.**, Falbén, J. K., Qian, C. S., & Sanborn, A. N. (2026). Generated Outcomes in Risky Choice Reveal Biased Sampling and Sequential Dependencies. *Communications Psychology*. In Press.
- 2 Sanborn, A. N., Zhu, J.-Q., Spicer, J., León-Villagrà, P., **Castillo, L.**, Falbén, J. K., Li, Y.-X., Tee, A., & Chater, N. (2025). Noise in Cognition: Bug or Feature? *Perspectives on Psychological Science*, 20(3), 572–589. <https://doi.org/10.1177/17456916241258951>
- 1 **Castillo, L.**, León-Villagrà, P., Chater, N., & Sanborn, A. N. (2024). Explaining the flaws in human random generation as local sampling with momentum. *PLOS Computational Biology*, 20(1), 1–24. <https://doi.org/10.1371/journal.pcbi.1011739>

Preprints

- 6 **Castillo, L.**, León-Villagrà, P., Falbén, J. K., Chater, N., & Sanborn, A. N. (2025). *People Need a Few Seconds to Be Random*. PsyArXiv. <https://doi.org/10.31234/osf.io/awg9j>
- 5 **Castillo, L.**, Li, Y.-X., & Sanborn, A. N. (2025). *The samplr package: A tool for modeling human cognition with sampling algorithms*. PsyArXiv. <https://doi.org/10.31234/osf.io/ax8hm>
- 4 Spicer, J., Li, Y.-X., **Castillo, L.**, Falben, J., Qian, S., & Sanborn, A. N. (2025). *Mental Sampling in Preferential Choice: Tracing the Process using Random Generation*. PsyArXiv. <https://doi.org/10.31234/osf.io/kzmb5>
- 3 Li, Y.-X., Falben, J., **Castillo, L.**, Spicer, J., Zhu, J.-Q., Qian, S., Chater, N., & Sanborn, A. N. (2025). *Does utility bias mental simulations of risky events?* PsyArXiv. <https://doi.org/10.31234/osf.io/awdr7>
- 2 Falbén, J., **Castillo, L.**, León-Villagrà, P., Chater, N., & Sanborn, A. N. (2024). *Biased mind or biased world? Assessing the accuracy of cultural beliefs that underlie social judgments*. PsyArXiv. <https://doi.org/10.31234/osf.io/5hdbp>
- 1 León-Villagrà, P., **Castillo, L.**, Falbén, J., Chater, N., & Sanborn, A. N. (2024). *Eliciting Beliefs with Random Generation Tasks*. <https://doi.org/10.31234/osf.io/687au>

Peer-reviewed Conference Proceedings Articles

- 4 Valentin, S.[†], **Castillo, L.**[†], Sanborn, A. N., & Lucas, C. G. (2024). Distinguishing Between Process Models of Causal Learning. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 46. <https://escholarship.org/uc/item/2p43j2cw>
- 3 Li, Y.-X., Falbén, J. K., **Castillo, L.**, Spicer, J., Zhu, J.-Q., Chater, N., & Sanborn, A. N. (2024). Probability, but not utility, influences repeated mental simulations of lotteries. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 46. <https://escholarship.org/uc/item/2ps2t4sz>
- 2 León-Villagrà, P., **Castillo, L.**, Chater, N., & Sanborn, A. N. (2022). Eliciting Human Beliefs using Random Generation. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 44. <https://escholarship.org/uc/item/88k526dj>
- 1 **Castillo, L.**, León-Villagrà, P., Chater, N., & Sanborn, A. N. (2021). Local Sampling with Momentum Accounts for Human Random Sequence Generation. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 43. <https://escholarship.org/uc/item/3gz154kg>

† = equal contribution

Conferences

Workshops

July 2026 **Annual Meeting of the Cognitive Science Society** *Autocorrelated Sampling in Cognition: Implementing MCMC Models and Fitting Them Without Likelihoods*

Conferences (continued)

Talks

July 2026	European Mathematical Psychology Group Meeting <i>Autocorrelated Sampling in Cognition: Implementing MCMC Models and Fitting Them Without Likelihoods</i>
July 2024	MathPsych <i>Characterizing People's Sampling Engines Using Random Generation</i>
August 2023	SPUDM <i>Production rate effects on random generation elicitation</i>
July 2023	MathPsych <i>Modelling speeded random generation as sampling for inference.</i>
July 2022	Virtual MathPsych <i>Using sampling algorithms to explain human random generation.</i>
October 2021	Computational Cognition Workshop <i>Human random generation as a locally-bound process.</i>

Poster Presentations

July 2024	Annual Meeting of the Cognitive Science Society <i>People Need About Five Seconds to be Random: Autocorrelated Sampling Algorithms Can Explain Why</i>
November 2021	Annual Meeting of the Psychonomic Society <i>Local sampling can explain human deviations from randomness.</i>
July 2021	Annual Meeting of the Cognitive Science Society <i>Local sampling with momentum accounts for human random sequence generation.</i>

Other

July 2026	Summer school in Cognitive Modeling & Simulation-Based Inference. Columbus OH, USA
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Reviews

Since	Venue	Count
2026	Cognition	1
2024	Journal of Experimental Psychology: General	2
2023	Journal of Experimental Psychology: Learning, Memory, and Cognition	1
2024	Cogsci Conference	16
2023	SPUDM Conference	5
2026	Psychological Science (Computational Reproducibility Check)	1

Misc.

Other Outputs

R Packages	Castillo, L., Li, Y.-X., & Sanborn, A. N. (2024). samplr: Compare Human Performance to Sampling Algorithms. https://CRAN.R-project.org/package=samplr
	Castillo, L., Li, Y.-X., & Sanborn, A. N. (2024). samplrData: Datasets from the SAMPLING Project. https://CRAN.R-project.org/package=samplrData .

Other University Involvement

2024 – present	Member of ReproducibiliTea Organizing Committee at UNIVERSITY OF WARWICK, UK (University-wide Open Research initiative)
2022 – 2023	Student Representative for Psychology PhD Students at UNIVERSITY OF WARWICK, UK

Misc. (continued)

- 2021 – 2023 **Teaching Assistant**, UNIVERSITY OF WARWICK, UK.
Modules: *Methods and Analysis in Behavioural Science* and
Bayesian Approaches in Behavioural Science
- 2019 – 2020 **Student Representative** for M.Sc. Cognitive and Decision Sciences cohort at UNIVERSITY COLLEGE LONDON, UK

Awards and Scholarships

- 2025 **Best PhD Student Paper** DEPARTMENT OF PSYCHOLOGY, UNIVERSITY OF WARWICK, UK.
Granted for Castillo et al. 2024.
- 2021 **Departmental PhD Scholarship** DEPARTMENT OF PSYCHOLOGY, UNIVERSITY OF WARWICK, UK
- 2020 **Best performing student** of the academic year for the MSc in Cognitive and Decision Sciences, UNIVERSITY COLLEGE LONDON, UK.

Skills

- Languages Spanish (mother tongue), Catalan, English (fluent), French, German (learning!)
- Coding git, Python, R, C++, $\text{\LaTeX}$, HTML, CSS, JavaScript, REACTJS.

References

Available upon request